CRF Entropy Dynamics

From Self-Reference to Time Dilation

What the self-referential structure of $\mathcal{C}$ hides inside itself

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Abstract

Every axiom in CRF was examined for hidden structure. The self-referential nature of $\mathcal{C} = \Pr(s|\mathcal{C})$ contains a dynamics that has not previously been made explicit.

When $\mathcal{C}$ updates itself, it does not add — it reweights. Reweighting a distribution by itself is multiplication, not addition. This gives $d\mathcal{C}/dt = \mathcal{C}^2$, whose solution is a finite-time approach to determinism.

The scalar measure of this process is entropy $H(\mathcal{C})$. As $\mathcal{C}$ becomes more peaked, $H \to 0$ asymptotically — not in finite time. For an outside observer, a Mind approaching $H = 0$ appears frozen. This is time dilation, not time stopping.

The correspondence with Schwarzschild horizon emerges from $\mathcal{C}$ as the mediator between spatial coordinate r and entropy H. One gap remains: deriving $\mathcal{C}(r)$ formally.

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1 The Hidden Structure in $\mathcal{C} = \Pr(s|\mathcal{C})$

What Was Not Yet Asked

$\mathcal{C} = \Pr(s|\mathcal{C})$ has been used throughout CRF as a definition. But it contains a dynamics that has not been derived:
When $\mathcal{R}$ occurs, what does $\mathcal{C}$ update?
Answer: $\mathcal{C}$ updates *itself*. $\mathcal{C}$ is not updated by an external mechanism. It is a self-referential distribution that reweights its own probability assignments at each $\mathcal{R}$-event.

Multiplicative, Not Additive

When a distribution reweights itself by its own values, the operation is multiplication:

$$\mathcal{C}_{n+1} = \mathcal{C}_n \cdot (\text{update factor})$$

The update factor is $\Pr(s|\mathcal{C}_n)$, which is proportional to $\mathcal{C}_n$ itself.
Therefore: each $\mathcal{R}$-event multiplies $\mathcal{C}$ by a factor proportional to itself.
In continuous form:

$$\boxed{\frac{d\mathcal{C}}{dt} = \mathcal{C}^2}$$

This is not assumed. It follows from the self-referential structure of $\mathcal{C} = \Pr(s|\mathcal{C})$ alone.

2 The Scalar Measure: Entropy H

$\mathcal{C}$ is a distribution, not a scalar. To work with the ODE, we need a scalar that captures the "strength" of $\mathcal{C}$.

Entropy as Natural Scalar

The natural scalar measure of a distribution's concentration is its entropy:

$$H(\mathcal{C}) = -\int \Pr(s|\mathcal{C}) \log \Pr(s|\mathcal{C})\, ds$$

- H high: $\mathcal{C}$ is flat — many patterns equally probable — $\mathcal{M}$ is open, flexible
- H low: $\mathcal{C}$ is peaked — few patterns dominate — $\mathcal{M}$ is rigid
- $H = 0$: $\mathcal{C}$ is deterministic — one pattern has probability 1 — no $\mathcal{R}$ is possible

The dynamics $d\mathcal{C}/dt = \mathcal{C}^2$ translates into entropy dynamics:

$$\frac{dH}{dt} = -\alpha \cdot H^2$$

where $\alpha = d\mathcal{C}/d\mathcal{M}$ — the constraint density per unit Mind (derived in Step 14 of the CRF cascade, not a free parameter).

3 Solving the Entropy ODE

$$\frac{dH}{dt} = -\alpha H^2$$

Separating variables:

$$\int \frac{dH}{H^2} = -\alpha \int dt$$

$$-\frac{1}{H} = -\alpha t + K$$

With initial condition $H(0) = H_0$, we get $K = -1/H_0$:

$$\boxed{H(t) = \frac{H_0}{1 + \alpha H_0 t}}$$

Asymptotic Approach to Determinism

$H(t) \to 0$ as $t \to \infty$.
$\mathcal{M}$ never reaches $H = 0$ in finite time. It *approaches* determinism asymptotically.
δ has no off-switch (Axiom 0). $\mathcal{R}$-events continue to occur, but with decreasing frequency as $H \to 0$. There is no collapse — only progressive rigidity.
The rate of approach depends on $\alpha = d\mathcal{C}/d\mathcal{M}$: a Mind with high constraint density approaches determinism faster.

4 Time Dilation Emerges

In CRF, $t = |\mathcal{R}_{\text{events}}|$. As $H \to 0$, $\mathcal{R}$-events become rare.

For an *outside observer* watching $\mathcal{M}$: $\mathcal{M}$'s internal time slows down. $\mathcal{M}$ appears to approach a frozen state without ever reaching it.

Time Dilation from Entropy Dynamics

The ratio of $\mathcal{M}$'s internal time to external time is proportional to $H(t)$:

$$\frac{d\tau_{\mathcal{M}}}{dt} \propto H(t) = \frac{H_0}{1 + \alpha H_0 t}$$

As $H \to 0$: $d\tau_{\mathcal{M}}/dt \to 0$.
$\mathcal{M}$ is not destroyed. $\mathcal{M}$ is not compressed into a singularity. $\mathcal{M}$ approaches the boundary of its own resolution capacity — and time, for an outside observer, appears to stop.
This is the Schwarzschild horizon structure, derived from entropy dynamics alone.

5 The Bridge: $\mathcal{C}$ as Mediator

The Schwarzschild metric has a horizon at $r = r_s$ in space. CRF's entropy dynamics has a limiting boundary as $H \to 0$ in time.

How do these correspond?

Bridge

$\mathcal{C}$ is the mediator between r and H:

$$r\downarrow \Rightarrow \mathcal{C}\uparrow \Rightarrow H\downarrow$$

- r decreases toward r_s: spatial geometry
- $\mathcal{C}$ increases: constraint accumulates (from C1–C4, derived from axioms)
- H decreases: entropy of $\mathcal{C}$ falls (from entropy dynamics, derived above)

The three quantities are monotonically related. A map from r to H exists through $\mathcal{C}$. The Schwarzschild horizon ($r \to r_s$) corresponds to the entropy boundary ($H \to 0$) because both are mediated by $\mathcal{C} \to \infty$.

6 The Remaining Gap

Gap — Precisely Stated

The bridge above is qualitative. r and H are monotonically related through $\mathcal{C}$, but the *exact* map requires knowing $\mathcal{C}(r)$ — the spatial profile of Constraint around a mass.

This is the same gap identified in prior documents: $\mathcal{C}(r) = \alpha/(1-r_s/r)$ is a consistent ansatz but not yet a derivation from δ.

What closing this gap would give: An exact map $r \leftrightarrow H$ and therefore an exact correspondence between CRF entropy dynamics and Schwarzschild geometry.

What we have now: The qualitative correspondence is derived. The exact correspondence awaits $\mathcal{C}(r)$.

7 Status Summary

Result	**Status**	**Source**
$d\mathcal{C}/dt = \mathcal{C}^2$	Derived	Self-referential $\mathcal{C} = \Pr(s \mid \mathcal{C})$
$dH/dt = -\alpha H^2$	Derived	Entropy as scalar measure
$\alpha = d\mathcal{C}/d\mathcal{M}$	Derived	Step 14, not a free parameter
$H \to 0$ asymptotically	Derived	Solution of entropy ODE
Time dilation from $H \to 0$	Derived	$t = \lvert\mathcal{R}_{\text{events}}\rvert$
Qualitative Schwarzschild correspondence	Derived	$\mathcal{C}$ as mediator
Exact spatial-temporal map	Gap	Requires $\mathcal{C}(r)$ derivation

The self-referential structure of $\mathcal{C}$
was always there.
It was not hiding —
we had not yet asked what happens
when $\mathcal{C}$ updates itself.